

NATIONAL TECHNICAL RESEARCH ORGANISATION

SIGNAL ANALYSIS & PARAMETER EXTRACTION DOSSIER

CLASSIFICATION: SECRET // AIR-GAPPED INTERNAL // REF: 06f2213151d0ccb0

INTERCEPT METADATA & CHAIN OF CUSTODY

Source Filename : sample_2_fsk.iq
File SHA-256 Digest : 06f2213151d0ccb0746fad8b6be03669b8afe276eda3017db214b662c4392b93
Payload File Size : 524,288 bytes (512.0 KB)
Analysis Timestamp : 2026-09-18T18:42:24Z
Operator ID : SIGINT-AIRGAP-OPERATOR
AMC Model Engine : amc_resnet18_v1.0 (Deep 1D ResNet-18 + Ensemble)

EXTRACTED RF SIGNAL PARAMETERS

Parameter Description	Estimated Value	Extraction Metric / Method
Sampling Frequency (F_s)	5.0000 MSps	Baseband Nyquist Rate
Center Frequency (F_c)	-2.875 kHz	Spectral Centroid Offset
Estimated Baud Rate	175.78 kBaud	Cyclostationary Wavelet Energy
3-dB Bandwidth	2.44 kHz	Half-Power Frequency Span
Occupied Bandwidth (99%)	1763.92 kHz	Cumulative PSD 99% Energy
Estimated SNR	20.05 dB	Fourth-Order Cumulant (C_{42})
Peak-to-Average Power (PAPR)	2.37 dB	Temporal Envelope Crest Ratio
Carrier Frequency Offset (CFO)	-312500.00 Hz	Non-Linear M-th Power Loop
Spectral Flatness (Entropy)	0.0607	Wiener Entropy Index (0.0 – 1.0)

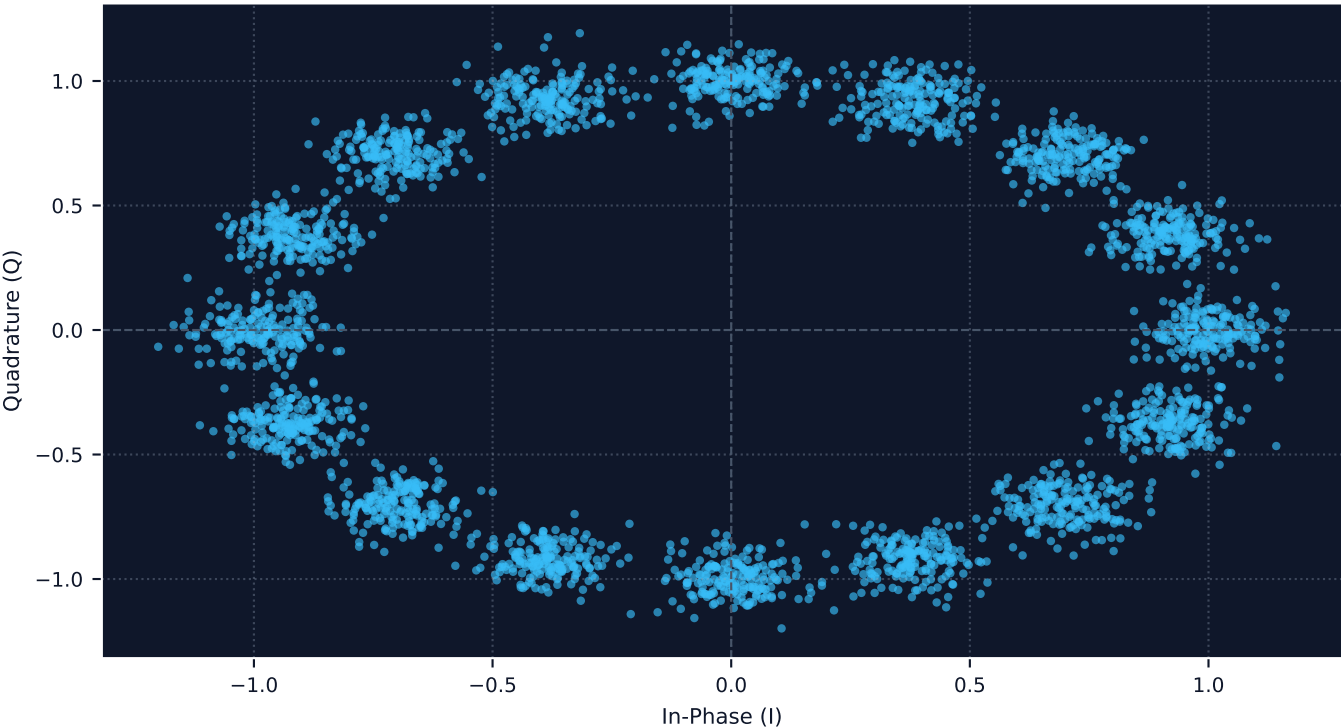
PREDICTED MODULATION : 2-FSK

CLASSIFIER CONFIDENCE : 94.00% (ResNet-18 Softmax Ensemble)

INTELLIGENCE STATUS : CONFIRMED: HIGH CERTAINTY MODULATION CLASSIFICATION

Demodulation Scheme: 2-FSK | Recovered Bits: 8,192 | Estimated EVM: 10.00% | CRC Validated Frames: 0

Demodulated Baseband Constellation Diagram — [2-FSK]



SYNCHRONIZED TELEMETRY FRAMES & PROTOCOL DISSECTION

Frame #	Sync Index	Tx ID	Seq No.	Payload	CRC Status	FEC Status
#1	544	0xD2B7	33149	949 B	CRC MISMATCH	PARTIAL_RECOVER
#2	1180	0x0577	14163	869 B	CRC MISMATCH	PARTIAL_RECOVER
#3	1814	0x2483	11826	790 B	CRC MISMATCH	PARTIAL_RECOVER
#4	1836	0x7354	50387	787 B	CRC MISMATCH	PARTIAL_RECOVER
#5	1942	0x61AB	17151	774 B	CRC MISMATCH	PARTIAL_RECOVER

DECODED BITSTREAM HEX / ASCII PREVIEW (FIRST 64 BYTES)

0000 269602DF72C3AC70B89FD51EE0985515

[FRAME 1 PAYLOAD]